

Request for Budgetary Technical and Commercial Proposal No. OVE75-RFQ-24

Subject: Leaching cascade reactors with agitators, Hastelloy G-35 / 904L (outside the Basic Engineering scope).

KVANT LLC (Russian Federation), within the Basic Engineering (FEED) stage of the construction project of a copper production plant — a roasting-leaching-electrowinning complex with a capacity of 75,000 tonnes of copper cathodes per year — kindly requests your budgetary technical and commercial proposal (budgetary quotation) for the item above. Project site: an operating metallurgical site in north-west Russia.

Summary of the item: 2 head repulpers of 10 m³ with maximum specific agitator power and recirculation + 4 main reactors of 90 m³ + a buffer reactor. Technical requirements are given in Annex 1 to this request.

Please include in your quotation:

1. budget price quoted separately on EXW (manufacturer's works) and DDP project site — an operating metallurgical site in north-west Russia (Incoterms 2020) terms, stating the currency and the validity period of the offer;
2. manufacturing lead time and estimated delivery time (weeks from order / advance payment);
3. overall dimensions and weights of the equipment unit and of the largest shipping blocks (for logistics studies);
4. reference list of similar supplies: product, process duty and parameters, year of supply, country;
5. materials of parts in contact with the process medium, and the scope of supply (instrumentation, spare parts, documentation);
6. utility consumption and installation requirements (if available).

This is a budgetary enquiry at the Basic Engineering (FEED) stage: it does not constitute an offer and creates no obligations for either party; detailed technical data sheets will be issued at the next project stage.

Reply deadline: we kindly ask you to submit your quotation within 3 (three) weeks from receipt of this request to stepan@kvantpro.com. Please confirm receipt of this request.

Confidentiality. This request and the attached technical materials are provided solely for the preparation of the quotation, are confidential, and shall not be disclosed to third parties, copied or published without the prior written consent of KVANT LLC. Submission of a quotation creates no legal obligations for either party.

Contact person: Stepan Kharkov, KVANT LLC, stepan@kvantpro.com.

Annex 1: Technical requirements for the requested item.

Annex 1. Technical Requirements

To Request No. OVE75-RFQ-24 · The values follow the Customer's input data and the Customer's Terms of Reference; Basic Engineering (FEED) stage, subject to refinement at the next project stage.

RFQ item	Leaching cascade reactors, Hastelloy G-35 / 904L (outside the Basic Engineering scope)
Summary	2 head repulpers of 10 m ³ with maximum specific agitator power and recirculation + 4 main reactors of 90 m ³ + a buffer reactor
Specific	Per the input data requirements, 316L material is not

requirements	acceptable — only Hastelloy G-35 or an equivalent; 904L is acceptable
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Equipment covered by this request:

1. **Head repulping reactors** Leaching (outside the Basic Engineering scope, for battery-limit coordination) · items 4.1.10, 4.1.11

Duty	Start of calcine leaching; maximum-intensity agitation, conical bottom, slurry recirculation or an air lift
Key parameters	Contact time 0.05 h; flow 124.7 m ³ /h; calculated volume 6.2 m ³
Quantity	2 in parallel (one backing up the other during accretion clean-out)
Materials	Hastelloy G-35 (Haynes Int.) or equivalent; 904L steel is acceptable
Dimensions, hold-up	V = 10.0 m ³ ; actual 16 m ³ ; margin factor 2.6

2. **Main leaching cascade reactors** Leaching (outside the Basic Engineering scope) · items 4.1.12–4.1.14

Duty	Main leaching of calcine in recycle solution from electrowinning
Key parameters	Contact time 2.0 h; flow 124.7 m ³ /h; calculated volume 249.5 m ³
Quantity	4
Materials	Hastelloy G-35 or 904L
Dimensions, hold-up	V = 90 m ³ ; actual 288 m ³ ; margin factor 1.2

3. **Buffer reactor** Leaching (outside the Basic Engineering scope) · item 4.1.15

Duty	Buffer ahead of the thickener; receiving slurry from the check filtration filter
Key parameters	Volume reserve 0.5 h; calculated volume 62.4 m ³
Quantity	1
Materials	Hastelloy G-35 or 904L
Dimensions, hold-up	V = 90 m ³ ; actual 72 m ³ ; margin factor 1.2

The quotation must confirm: process medium, temperature, pressure, materials of wetted parts, capacity, dimensions and weight, power consumption, manufacturing lead time, delivery terms.

预算性技术商务报价询价函 编号 OVE75-RFQ-24

询价主题：浸出串联反应槽（带搅拌器），Hastelloy G-35 / 904L（基础设计范围外）。

KVANT LLC（俄罗斯）正在开展一座铜冶炼厂建设项目的基础设计（FEED）——“焙烧—浸出—电积”联合工艺，年产阴极铜 75,000 吨——特此请贵方就上述设备提交预算性技术商务报价。项目场地：俄罗斯西北部一座现役冶金厂区。

设备简述：2 台 10 m³ 头部再制浆槽（搅拌器比功率最大并带再循环）+ 4 台 90 m³ 主反应槽 + 1 台缓冲槽。技术要求见本询价函附件 1。

请在报价中包含以下内容：

1. 预算价格——分别按 EXW（制造厂交货）和 DDP 项目场地（俄罗斯西北部现役冶金厂区）条款报价（Incoterms 2020），并注明币种及报价有效期；
2. 制造周期及预计交货期（自下单/预付款之日起，按周计）；
3. 外形尺寸与重量：单台设备及最大发运件的尺寸和重量（用于物流研究）；
4. 类似供货业绩表：产品、工艺介质及参数、供货年份、国家；
5. 与工艺介质接触部件的材质，以及供货范围（仪表、备品备件、文件资料）；
6. 公用工程消耗及安装要求（如有资料）。

本询价为基础设计（FEED）阶段的预算性询价：不构成要约，对双方均不产生任何义务；详细技术数据表将在项目下一阶段发出。

回复期限：请自收到本询价函之日起 3（三）周内将报价发送至 stepan@kvantpro.com。请确认收到本询价函。

保密条款：本询价函及所附技术资料仅供编制报价使用，属于保密信息，未经 KVANT LLC 事先书面同意，不得向第三方披露、复制或发表。提交报价不构成双方的法律义务。

联系人：Stepan Kharkov, KVANT LLC, stepan@kvantpro.com。

附件 1：本询价设备的技术要求。

附件 1. 技术要求

对应询价函编号 OVE75-RFQ-24 · 参数依据业主原始数据及设计任务书；基础设计（FEED）阶段，下一阶段可能进一步细化。

询价设备	浸出串联反应槽，Hastelloy G-35 / 904L（基础设计范围外）
简述	2 台 10 m ³ 头部再制浆槽（搅拌器比功率最大并带再循环）+ 4 台 90 m ³ 主反应槽 + 1 台缓冲槽
特殊要求	按原始数据要求，不允许使用 316L 材质——仅限 Hastelloy G-35 或同等材质；允许使用 904L

本询价涵盖的设备清单：

1. **头部再制浆反应槽** 浸出（基础设计范围外，用于界区衔接）· 位号 4.1.10、4.1.11

用途	焙砂浸出的起始工序；最大强度搅拌，锥形槽底，矿浆再循环或空气气升
主要参数	接触时间 0.05 h；流量 124.7 m ³ /h；计算容积 6.2 m ³
数量	2 台并联（清理结壳期间一台替代另一台运行）
材质	Hastelloy G-35（Haynes Int.）或同等材质；允许使用 904L

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尺寸、持液量	$V = 10.0 \text{ m}^3$; 实际 16 m^3 ; 裕量系数 2.6

2. 浸出串联主反应槽 浸出 (基础设计范围外) · 位号 4.1.12-4.1.14

用途	焙砂在电积返回液中的主浸出
主要参数	接触时间 2.0 h ; 流量 $124.7 \text{ m}^3/\text{h}$; 计算容积 249.5 m^3
数量	4 台
材质	Hastelloy G-35 或 904L
尺寸、持液量	$V = 90 \text{ m}^3$; 实际 288 m^3 ; 裕量系数 1.2

3. 缓冲反应槽 浸出 (基础设计范围外) · 位号 4.1.15

用途	浓密机前的缓冲槽 ; 接收检查过滤滤机的矿浆
主要参数	容积储备 0.5 h ; 计算容积 62.4 m^3
数量	1 台
材质	Hastelloy G-35 或 904L
尺寸、持液量	$V = 90 \text{ m}^3$; 实际 72 m^3 ; 裕量系数 1.2

报价中必须确认：工艺介质、温度、压力、接触部件材质、生产能力、尺寸与重量、能耗、制造周期、交货条件。